

DevSecOps Multi-Engine Pack - Hands Build Report (D1 to D6)

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Product: Sentinel Stacks
Role agent: DevSecOps
Tool ID: `scan_devsecops_pack`
Module: `ai_devsecops_pack.py`
Final version: `0.6.0-d6`
Status: Hands **COMPLETE** (10 / 10 engines active)
Origin tip: `07397d5` ON `main` (<https://github.com/ibkbalo/secops-pipeline-lab>)
Lab path: `C:\DevSecOps-Lab\secops-pipeline-lab`
Report date: 2026-07-29
Scope of this document: DevSecOps **hands** only (deterministic scanners). Brain / Face / FIX_MAP expansion are out of scope for this ship.

1. Executive summary

Between **2026-07-27** and **2026-07-28**, Sentinel Stacks closed the DevSecOps role gap with a **multi-engine enterprise pack** - not a toy "18-check" audit. The pack was delivered in six sequential phases (D1->D6), each dual-verified (local smoke + Cursor MODE B) before commit and push to `origin/main`.

Metric	Final (D6)
Version	<code>0.6.0-d6</code>
Engines registered	10
Engines active	10
Engines stub	0
Pack complete	100%
<code>pack_hands_complete</code>	true
Vuln mock findings	62
Clean mock findings	0 (<code>checks_passed = 10</code>)
Finding ID scheme	<code>DEVSEC - {CODE} - {NNN}</code>

Bottom line: DevSecOps hands are done. One pack run answers secrets, SCA, containers, IaC, SAST, CI/CD, supply chain, policy-as-code, repo governance, and release gates. Next product step (when directed): expand Remediation **FIX_MAP** for `DEVSEC - *` IDs, then role **brain**.

2. Product context

2.1 Four role agents (Sentinel Stacks)

#	Role	Hands status at D6 close
1	Security Engineer	Perimeters + remediation kits (prior ship)
2	Cloud Security Engineer	AWS + Azure auditors (prior ship)
3	DevSecOps	This pack - complete at D6
4	AI Security Engineer	Still open

2.2 Shared spine

- **TOOL STANDARDS v1.0** report contract (`tool_id` , `version` , `execution` , `summary` , `findings[]` , `metadata`)
- **Remediation Sentinel** (`remediate_findings_hardening_kit` v1.1.0) - `FIX_MAP` currently perimeter/cloud-heavy; DevSecOps IDs are next
- **tool_registry.json** + deterministic **mock fixtures** for offline demos / recruiter smoke
- Product order: **Hands -> Brain -> Face** (this report is Hands)

2.3 Design principles (locked early)

1. **Enterprise bar, no floor ceiling** - capacity grows by engine; never freeze at an "18-check" demo.
2. **Deterministic mock path** - `#mock` + vulnerable/clean multi-engine fixtures always work offline.
3. **Optional live backends** when installed (gitleaks, Trivy, Checkov, semgrep, syft, ...) without hard dependency.
4. **Stable finding IDs** - `DEVSEC-{ENGINE}-{NNN}` forever; NNN grows without renumber chaos.
5. **Hands only** - no LLM brain loop inside the pack; findings must be machine-actionable later.
6. **Dual gate before ship** - local smoke PASS + Cursor MODE B PASS -> commit/push only on `COMMIT_OK` .

3. Architecture

3.1 Module surface

```
scan_devsecops_pack (ai_devsecops_pack.py)
|
+-- ENGINE_REGISTRY (10 engines: active|stub)
+-- Backend probes (which CLI tools exist)
+-- Fixture load (mock JSON) --or-- live repo path
+-- Per-engine workers (_engine_*)
+-- Merge + severity / risk domain scores
+-- TOOL_STANDARDS JSON report
```

3.2 Engine registry (final)

Key	Code	Phase activated	What it answers
secrets	SEC	D2	Are credentials leaking in repo/CI?
cicd	CICD	D2	Is the pipeline identity & gate set safe?
sca	SCA	D3	Are dependencies CVE-ridden / unpinned?
container	CTR	D3	Are images / Dockerfiles built securely?
iac	IAC	D4	Is infra-as-code misconfigured?
policy	POL	D4	Is policy-as-code enforced?
supply_chain	SC	D5	SBOM, signing, SLSA, dep-review?
repo_gov	GOV	D5	CODEOWNERS, SECURITY.md, workflow defaults?
release	REL	D5	Provenance, retention, prod approvals?
sast	SAST	D6	Source dangerous sinks + CI SAST gate?

3.3 Finding ID scheme

```
DEVSEC- {ENGINE_CODE} - {NNN}

ENGINE_CODE in {
  SEC, SCA, CTR, IAC, SAST, CICD, SC, POL, GOV, REL
}
```

Example: DEVSEC-SAST-001 , DEVSEC-CICD-014 .

3.4 Modes of operation

Mode	Invocation (examples)	Purpose
Vulnerable mock	<code>python ai_devsecops_pack.py mock</code>	Deterministic failure profile (demo / CI bar)
Clean mock	<code>python ai_devsecops_pack.py .</code> <code>mock_devsecops_clean.json</code>	Proves engines screw only when evidence exists
Live / embedded	<code>python ai_devsecops_pack.py .</code> or path	Repo walk + embedded patterns + optional CLIs

Windows hygiene (locked): write JSON via `out-File -Encoding utf8` , load with `utf-8-sig` (default `>` is UTF-16 and breaks `json.load`).

3.5 Optional live backends (probe at runtime)

Tool	Engines that can use it
gitleaks	secrets
actionlint	cicd
trivy	sca, container, iac
checkov	iac, policy
semgrep	sast
syft / gype	supply_chain, sca
embedded	always available for all 10

If a CLI is missing, pack still runs; mock fixtures + embedded logic produce the deterministic bar.

3.6 Files that make up the product surface

File	Role
ai_devsecops_pack.py	Pack facade + all engines
tool_registry.json	Catalog entry scan_devsecops_pack
mock_devsecops_vulnerable.json	Multi-engine failure fixture
mock_devsecops_clean.json	Multi-engine pass fixture

Smoke artifacts (.tmp_*.json) are **never** committed.

4. Delivery workflow (every phase)

```
Sandbox implement -> smoke (vuln + clean)
->
Handoff zip / patch (VERIFY_Dx.md)
->
Windows lab copy (unzip -> 4 product files)
->
Local smoke (utf8 Out-File)
->
Cursor MODE B (read-only verify)
->
COMMIT_OK -> git add (4 files only) -> commit -> push origin/main
->
Optional remote tip confirm
```

This process shipped **D1 through D6** without scope creep into brain/Face.

5. Phase-by-phase build log

5.1 Cohort rollup

Phase	Commit	Date (author)	Version	Engines activated	Active	Stub	pack%	Vuln findings	Clean
D1	dcf54c9	2026-07-27	skeleton	none (registry only)	0	10	0	0 (by design)	0
D2	840ac28	2026-07-27	0.2.0-d2	secrets, cicd	2	8	20	19	0 / passed 2
D3	109c534	2026-07-28	0.3.0-d3	+ sca, container	4	6	40	33	0 / passed 4
D4	362dd70	2026-07-28	0.4.0-d4	+ iac, policy	6	4	60	43	0 / passed 6
D5	168fc4b	2026-07-28	0.5.0-d5	+ supply_chain, repo_gov, release	9	1	90	56	0 / passed 9
D6	07397d5	2026-07-28	0.6.0-d6	+ sast	10	0	100	62	0 / passed 10

Cumulative finding math (vulnerable mock):

```
D2:  5 SEC + 14 CICD                = 19
D3:  19 + 6 SCA + 8 CTR              = 33
D4:  33 + 5 IAC + 5 POL              = 43
D5:  43 + 5 SC + 4 GOV + 4 REL       = 56
D6:  56 + 6 SAST                     = 62
```

5.2 D1 - Pack skeleton (dcf54c9)

Question answered: *What is a multi-engine DevSecOps hands pack on TOOL_STANDARDS, before real checks?*

Shipped:

- `ai_devsecops_pack.py` skeleton (~706 lines): engine registry, ID codes, backend detect shell, merge runner, domain scoring shell
- `mock_devsecops_vulnerable.json` / `mock_devsecops_clean.json` multi-section fixture schema
- Registry entry `scan_devsecops_pack` with 10 engine keys, all **stub**
- Commit title: `feat(devsecops): D1 multi-engine pack skeleton (enterprise bar)`

Smoke bar: findings **0 / 0** by design - stubs emit no noise.

Why it mattered: locked architecture so later phases only *activate* engines instead of rewriting the facade.

5.3 D2 - Secrets + CI/CD (840ac28 , 0.2.0-d2)

Questions answered:

- Secrets: *Is identity material leaking in tracked files or CI plain env?*
- CI/CD: *Is the pipeline over-privileged, unpinned, ungated?*

Coverage (vulnerable mock):

Code	Count	Themes
SEC	5	API key / AWS AKIA / GitHub PAT in tracked files; DB_PASSWORD plaintext in CI; secret scanning off
CICD	14	Overbroad permissions, unpinned Actions, curl bash install, plain secrets in deploy env, force-push on main, no PR/status/CODEOWNER gates

Live backends (optional): gitleaks, actionlint.

Bar: vuln **19** | clean **0** | pack **20%**.

Commit: feat(devsecops): D2 activate secrets + cicd engines (0.2.0-d2).

5.4 D3 - SCA + Container (109c534, 0.3.0-d3)

Questions answered:

- SCA: *What we build with - known CVEs, lockfiles, automation?*
- Container: *What we ship as - root, latest tags, HEALTHCHECK, image OS CVEs?*

Coverage (new on vuln bar):

Code	Count	Themes
SCA	6	Django JSONField SQLi CVE, lodash template cmd inject, PyYAML full_load RCE; missing lockfiles; Dependabot off
CTR	8	Floating base, run-as-root, no HEALTHCHECK, broad COPY, image OS critically/high CVEs

Live backends (optional): Trivy fs / dockerfile / image.

Bar: vuln **33** | clean **0** | pack **40%**.

Commit: feat(devsecops): D3 activate SCA + container engines (0.3.0-d3).

5.5 D4 - IaC + Policy-as-Code (362dd70, 0.4.0-d4)

Questions answered:

- IaC: *Do Terraform/K8s blueprints open the world or skip limits?*
- Policy: *Is OPA/Conftest/admission feedback in the SDLC loop?*

Coverage (new):

Code	Count	Themes
IAC	5	SSH SG 0.0.0.0/0, S3 no public block, K8s privilege escalation / no limits, IaC scan gate off
POL	5	No OPA, no Conftest, admission off, explicit violation, PaC not required

Live backends (optional): Trivy config, Checkov.

Bar: vuln **43** | clean **0** | pack **60%**.

Commit: feat(devsecops): D4 activate IaC + policy engines (0.4.0-d4).

5.6 D5 - Supply chain + Repo gov + Release (168fc4b , 0.5.0-d5)

Questions answered:

- Supply chain: *SBOM, signatures, SLSA, PR dep-review, install-pipe hygiene?*
- Repo gov: *CODEOWNERS, SECURITY.md, default workflow token scope?*
- Release: *Retention, provenance, environment protection, prod approval?*

Coverage (new):

Code	Count	Themes
SC	5	No SBOM, unsigned images, SLSA 0, no dep-review on PR, remote install pipe
GOV	4	Missing CODEOWNERS / SECURITY.md, no path reviewers, read-write default permissions
REL	4	1-day retention, no attestations, unprotected env, no manual prod approval

Live backends (optional): syft (SBOM-oriented path).

Bar: vuln **56** | clean **0** / checks_passed **9** | pack **90%**.

Stub remaining: sast only.

Commit: feat(devsecops): D5 activate supply_chain + release + repo_gov (0.5.0-d5) .

5.7 D6 - SAST + pack hands complete (07397d5 , 0.6.0-d6)

Question answered: *Is source full of classic sinks, and is SAST actually required in CI?*

Coverage (new):

Code	Count	Themes
SAST	6	sql-injection-string-format, hardcoded-jwt-secret, command-injection-os-system, xss-unescaped-template; sast_in_ci=false gate; dangerous sinks unreviewed

Implementation notes:

- Fixture-driven issues + process gates
- Optional **semgrep** live when installed
- Embedded Python danger-pattern fallback for live mode
- Registry: pack_phase: D6 , pack_hands_complete: true , complete_pct: 100
- Readiness next_phase : *FIX_MAP expand for DEVSEC-* IDs -> role brain after all-role hands*

Bar: vuln **62** | clean **0** / checks_passed **10** | pack **100%** | active **10** | stub **0**.

Commit message:

```
feat(devsecops): D6 activate SAST engine - pack hands complete (0.6.0-d6)
```

Dual gate outcomes:

- Local Windows smoke: PASS (62 / 0 / 100% / hands_complete True)
- Cursor MODE B: **VERDICT PASS** -> **COMMIT_OK**
- Push: 168fc4b..07397d5 main -> main

6. Full vulnerable fixture catalog (D6 tip)

Severity mix at D6: **critical 20 | high 32 | medium 10 | low 0 | info 0** (62 total).

6.1 Secrets (SEC) - 5

ID	Severity	Title
DEVSEC-SEC-001	high	Secret in tracked file: generic-api-key
DEVSEC-SEC-002	critical	Secret in tracked file: aws-access-key-id
DEVSEC-SEC-003	critical	Secret in tracked file: github-pat
DEVSEC-SEC-004	critical	CI plaintext secret env: DB_PASSWORD
DEVSEC-SEC-005	high	Repository secret scanning disabled

6.2 CI/CD (CICD) - 14

ID	Severity	Title
DEVSEC-CICD-001	critical	Overbroad workflow permissions: ci.yml
DEVSEC-CICD-002	high	Unpinned Actions in ci.yml
DEVSEC-CICD-003	high	No security jobs in pipeline: ci.yml
DEVSEC-CICD-004	critical	Dangerous CI pattern curl-bash-install: ci.yml
DEVSEC-CICD-005	critical	Overbroad workflow permissions: deploy.yml
DEVSEC-CICD-006	high	Unpinned Actions in deploy.yml
DEVSEC-CICD-007	critical	Secrets exposed via plain env: deploy.yml
DEVSEC-CICD-008	high	No security jobs in pipeline: deploy.yml
DEVSEC-CICD-009	high	Dangerous CI pattern unpinned-action: deploy.yml
DEVSEC-CICD-010	high	Dangerous CI pattern overbroad-permissions: deploy.yml
DEVSEC-CICD-011	critical	Force push allowed on main
DEVSEC-CICD-012	high	PR review not required on main
DEVSEC-CICD-013	high	No required status checks on main
DEVSEC-CICD-014	medium	CODEOWNER review not required on main

6.3 SCA - 6

ID	Severity	Title
DEVSEC-SCA-001	critical	CVE-2019-14234: Django SQL injection via JSONField/HStoreField
DEVSEC-SCA-002	high	CVE-2021-23337: Command injection in lodash template
DEVSEC-SCA-003	critical	CVE-2020-14343: PyYAML arbitrary code execution via full_load
DEVSEC-SCA-004	high	Missing lockfile for requirements.txt
DEVSEC-SCA-005	high	Missing lockfile for package.json
DEVSEC-SCA-006	medium	Dependency update automation disabled

6.4 Container (CTR) - 8

ID	Severity	Title
DEVSEC-CTR-001	high	Floating/unpinned base image: Dockerfile
DEVSEC-CTR-002	critical	Container runs as root: Dockerfile
DEVSEC-CTR-003	medium	Missing HEALTHCHECK: Dockerfile
DEVSEC-CTR-004	critical	Container runs as root: services/api/Dockerfile
DEVSEC-CTR-005	medium	Missing HEALTHCHECK: services/api/Dockerfile
DEVSEC-CTR-006	high	Broad build context copy: services/api/Dockerfile
DEVSEC-CTR-007	critical	Image OS critical vulns (3): enterprise-app:latest
DEVSEC-CTR-008	high	Image OS high vulns (12): enterprise-app:latest

6.5 IaC - 5

ID	Severity	Title
DEVSEC-IAC-001	critical	Terraform: sg-ssh-world-open
DEVSEC-IAC-002	high	Terraform: s3-no-public-block
DEVSEC-IAC-003	high	Kubernetes: privilege-escalation-allowed
DEVSEC-IAC-004	medium	Kubernetes: no-resource-limits
DEVSEC-IAC-005	high	IaC security scanning disabled

6.6 Policy (POL) - 5

ID	Severity	Title
DEVSEC-POL-001	high	No OPA/Rego policies present
DEVSEC-POL-002	medium	Conftest / policy test harness missing
DEVSEC-POL-003	high	Cluster admission controls disabled
DEVSEC-POL-004	high	Policy violation: deny-privileged-containers
DEVSEC-POL-005	medium	Policy-as-code not required in SDLC

6.7 Supply chain (SC) - 5

ID	Severity	Title
DEVSEC-SC-001	high	SBOM not published with releases
DEVSEC-SC-002	critical	Container images not signed
DEVSEC-SC-003	high	SLSA level too low (0)
DEVSEC-SC-004	high	Dependency review missing on PRs
DEVSEC-SC-005	critical	Dangerous remote install pipe: ci.yml

6.8 Repo governance (GOV) - 4

ID	Severity	Title
DEVSEC-GOV-001	high	CODEOWNERS missing
DEVSEC-GOV-002	medium	SECURITY.md missing
DEVSEC-GOV-003	medium	No path-based required reviewers
DEVSEC-GOV-004	critical	Default workflow permissions too broad (read-write)

6.9 Release (REL) - 4

ID	Severity	Title
DEVSEC-REL-001	high	Artifact retention too short (1 days)
DEVSEC-REL-002	high	Release provenance attestations missing
DEVSEC-REL-003	critical	Deployment environment protection disabled
DEVSEC-REL-004	high	Production deploy lacks manual approval

6.10 SAST - 6

ID	Severity	Title
DEVSEC-SAST-001	critical	SAST: sql-injection-string-format
DEVSEC-SAST-002	high	SAST: hardcoded-jwt-secret
DEVSEC-SAST-003	critical	SAST: command-injection-os-system
DEVSEC-SAST-004	high	SAST: xss-unescaped-template
DEVSEC-SAST-005	high	SAST not required in CI pipeline
DEVSEC-SAST-006	medium	Dangerous sinks present without review coverage

7. Clean fixture contract

Clean profile must stay **void of findings** while proving engines ran:

Field	Expected (D6)
version	0.6.0-d6
execution.status	success
summary.total_findings	0
summary.checks_passed	10
summary.engines_active	10
summary.engines_stub	0
summary.pack_complete_pct	100
metadata.pack_readiness.pack_hands_complete	true

This is the anti-false-positive gate: recruiting demos and CI regression use the same pair.

8. TOOL_STANDARDS report shape (excerpt)

Top-level keys on every pack run:

```
{
  "tool_id": "scan_devsecops_pack",
  "version": "0.6.0-d6",
  "execution": {
    "timestamp": "...",
    "duration_seconds": 0.0,
    "target": "...",
    "status": "failed|success",
    "mode": "mock|live",
    "error": null
  },
  "summary": {
    "total_findings": 62,
    "critical": 20,
    "high": 32,
    "medium": 10,
    "low": 0,
    "info": 0,
    "risk_score": 0,
    "checks_run": 10,
    "checks_passed": 0,
    "engines_run": 10,
    "engines_active": 10,
    "engines_stub": 0,
    "pack_complete_pct": 100,
    "domain_scores": { }
  },
  "findings": [ { "id": "DEVSEC-...", "title": "...", "severity": "...", ... } ],
  "metadata": {
    "domain": "devsecops",
    "pack_phase": "D6",
    "pack_readiness": {
      "phase": "D6",
      "complete_pct": 100,
      "pack_hands_complete": true,
      "next_phase": "FIX_MAP expand for DEVSEC-* IDs -> role brain after all-role hands"
    },
    "backends": { },
    "id_scheme": "DEVSEC-{ENGINE}-{NNN}",
    "engine_codes": { }
  }
}
```

Findings intentionally **do not** use a separate `finding_type` field - remediation keys off stable `id` + title/severity (same spine as perimeter/cloud).

9. Registry entry (D6)

Significant `tool_registry.json` fields for `scan_devsecops_pack`:

Field	Value
version	0.6.0-d6
pack_phase	D6
pack_hands_complete	true
role	devsecops
domain	devsecops
module_file	ai_devsecops_pack.py
entry_point	run
active_engines	all 10 keys
engines_status.*	all active
Description (short)	Enterprise DevSecOps multi-engine pack - D6 COMPLETE: all 10 engines active (SAST closed). Next: DEVSEC FIX_MAP + role brain.

10. Commit ledger (main)

Hash	Subject	Diff sense
dcf54c9	feat(devsecops): D1 multi-engine pack skeleton (enterprise bar)	+1128 (pack + both mocks + registry)
840ac28	feat(devsecops): D2 activate secrets + cicd engines (0.2.0-d2)	+513 pack/registry
109c534	feat(devsecops): D3 activate SCA + container engines (0.3.0-d3)	+664 pack/vuln/registry
362dd70	feat(devsecops): D4 activate IaC + policy engines (0.4.0-d4)	+491 pack/mocks/registry
168fc4b	feat(devsecops): D5 activate supply_chain + release + repo_gov (0.5.0-d5)	+485 pack/registry
07397d5	feat(devsecops): D6 activate SAST engine - pack hands complete (0.6.0-d6)	+280 pack/mocks/registry

Tip confirmation:

```
07397d5 feat(devsecops): D6 activate SAST engine - pack hands complete (0.6.0-d6)
168fc4b feat(devsecops): D5 activate supply_chain + release + repo_gov (0.5.0-d5)
362dd70 feat(devsecops): D4 activate IaC + policy engines (0.4.0-d4)
```

```
git show HEAD:ai_devsecops_pack.py -> VERSION = "0.6.0-d6".
```

11. How to re-verify on Windows

```
cd C:\DevSecOps-Lab\secops-pipeline-lab

python ai_devsecops_pack.py mock 2>&1 | Out-File -Encoding utf8 .tmp_d6_vuln.json
python ai_devsecops_pack.py . mock_devsecops_clean.json 2>&1 | Out-File -Encoding
    utf8 .tmp_d6_clean.json

python -c "import json
for p in ['.tmp_d6_vuln.json', '.tmp_d6_clean.json']:
    r=json.load(open(p,encoding='utf-8-sig'))
    print(p, r['version'], r['execution']['status'],
          r['summary']['total_findings'], r['summary']['pack_complete_pct'],
          r['summary']['engines_active'], r['summary'].get('checks_passed'),
          r['summary']['engines_stub'])
    print([f['id'] for f in r['findings'] if f['id'].startswith('DEVSEC-SAST-')])
    print('hands_complete', (r.get('metadata') or {}).get('pack_readiness',
    {}).get('pack_hands_complete'))"
```

Do **not** `git add .tmp_*.json`.

12. Lessons coined during the build

1. **Stubs first (D1), activate later** - saved redesign cost every phase.
2. **Same four product files every handoff** - pack, registry, vuln mock, clean mock.
3. **PowerShell \$dtl / \$dst same session** - empty vars -> PathNotFound on Expand/Copy.
4. **Zip must actually land in Downloads** - D6 first unzip failed because only d3-d5 zips were saved.
5. **UTF-8 Out-File + utf-8-sig load** - mandatory on Windows for smoke JSON.
6. **Cursor MODE B before every commit** - kept history clean, prevented half-ships.
7. **Mock is not empty live** - running without mock uses embedded/live path; fixture bar only with `mock` or explicit clean JSON.

13. What this unlocks next

Priority	Work	Depends on hands pack
1	FIX_MAP entries for all DEVSEC-* IDs (Remediation Sentinel)	D6 complete [OK]
2	DevSecOps role brain (agentic loop over pack tools)	FIX_MAP + hands
3	AI Security Engineer hands pack	Independent parallel role
4	Face / dashboard scorecard binding	All role hands preferred

Pack readiness itself says:

```
next_phase : FIX_MAP expand for DEVSEC-* IDs -> role brain after all-role hands
```

14. One-page scoreboard

```
DevSecOps Hands Pack - CLOSED
-----
tool_id ..... scan_devsecops_pack
version ..... 0.6.0-d6
phase ..... D6
engines ..... 10 active / 0 stub
pack_complete_pct .... 100
hands_complete ..... true
vuln findings ..... 62
clean findings ..... 0
origin/main tip ..... 07397d5
repo ..... ibkballo/secops-pipeline-lab
role ..... DevSecOps (of 4 Sentinel agents)
next ..... FIX_MAP(DEVSEC-*) -> brain (on request)
```

15. Document control

Field	Value
Title	DevSecOps Multi-Engine Pack - Hands Build Report (D1 -> D6)
Classification	Internal product ship record / hands artifact
Formats	Markdown (this file)
Source of truth	origin/main @ 07397d5
Authoring agent	HackerAI build session (founder's authorized lab)
Audience	Founder, Cursor reviewers, future brain authors, recruiters (product layer)

End of report.